

The Economics of Superstars: Music Sales and Concerts

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Problem 10.4 [Applied]. Questions only. Part I uses consumer theory from Chapter 1; Parts II through IV combine cost minimization and monopoly pricing. This handout uses the same questions as the textbook.

Taylor S. is a stylized superstar whose fans consume music services and concerts. This is the graduate version of the music-and-concerts case, with explicit preferences, technology, and market assumptions. All numbers are hypothetical; the model is not an empirical description of Taylor Swift's contracts. Quantities are divisible, and the mass of identical fans is one.

Part I. Fan demand

Write q_M, q_C for music and concert services, the two coordinates of consumption. A fan has wealth $w > 0$, prices $p_M, p_C > 0$, and utility $u(q_M, q_C) = (\frac{1}{2}q_M^\rho + \frac{1}{2}q_C^\rho)^{1/\rho}$, where $0 < \rho < 1$. The letters M, C are local commodity labels.

(a) Formulate the budget problem and derive demands using $\sigma = 1/(1 - \rho)$. For the remaining parts set $\rho = 1/2$, $w = 1$, and $p_M = 1$. Explain why music demand changes when the concert price changes.

Part II. Production

For $j \in \{M, C\}$ the artist's technology is $y_j = A_j \sqrt{n_j}$, where $n_j \geq 0$ is labor and $A_j > 0$ productivity. Labor costs $\nu > 0$ per hour, distinct from fan wealth w . The processes are independent. This stylized music-service cost does not describe the cost of creating an original recording as a function of its streams.

(b) Derive $C_j(\nu, y_j)$, average cost, and marginal cost. Explain decreasing returns. Set $A_M = A_C = \nu = 1$ in the numerical parts.

Part III. A share of volume and receipts

An exogenous $\tau \in (0, 1]$ is the fraction of music consumption attributable to this artist. Her output is $y_M = \tau q_M$ and her music receipts are $\tau p_M q_M$. Concert output is $y_C = q_C$. She takes p_M, τ as given and chooses $p_C > 0$. There is no shared labor constraint or fixed cost.

(c) Write total revenue and profit after substituting demand and costs. Distinguish a revenue derivative with respect to ticket price from marginal revenue with respect to output. Account for changes in both revenue streams and production costs.

(d) Derive the first-order condition for the optimal concert price. Examine profit as $p_C \downarrow 0$ and $p_C \rightarrow \infty$. Establish global optimality and uniqueness, analytically or with a justified numerical method accounting for every stationary point and boundary.

(e) Compare $\tau = 1$ and $\tau = 1/2$. In a table report optimal concert prices, both fan quantities, both delivered outputs, revenues, costs, and profits. State numerical accuracy. Does the concert price rise or fall when the music share falls?

(f) Use the first-order condition or implicit differentiation to analyze the price response for $\tau \in [1/2, 1]$. Interpret the roles of substitution and production costs. Do not assume the same sign for every technology or parameter value.

Part IV. A royalty share

Now τ changes only the fraction of music revenue paid to the artist: music output is $y_M = q_M$, while receipts remain $\tau p_M q_M$. All other assumptions are unchanged.

(g) Write the revised profit function and repeat the comparison between $\tau = 1$ and $\tau = 1/2$. Identify the changed cost channel and determine whether the earlier price response survives.

(h) Explain what evidence distinguishes a fall in music volume from a fall in the royalty fraction. Discuss the roles of CES substitution, decreasing returns, and control of concert prices. Contrast the artist's decision with price-taking supply in the main chapter.